

Week 3 — Requirements Engineering & Mini-Capstone

"A PRD is a contract between the customer's problem and the engineering work that solves it. If either side cannot read it, the contract is void."

Week 3 is the **first capstone week** of the academy. Triads convert their Week-2 strategy package — North Star, design principles, journey map, top feature concepts — into a single **Technical PRD** that an engineering lead would accept as input to scoping. The PRD ships Friday after a structured peer-review cycle.

This week is **explicitly non-AI**. The discipline of writing a PRD by hand, with your triad, is the muscle being built. AI returns in Week 4.

Learning outcomes

By Friday afternoon, each participant can:

1. Convert a customer problem into a structured **Technical PRD** with context, problem framing, goals, scope, AC, NFRs, risks, and open questions.
2. Write **granular, testable Acceptance Criteria** in Given/When/Then form that an engineer can implement against without a clarifying call.
3. Document **Non-Functional Requirements** across performance, security, accessibility, observability, and compliance — and defend each as a constraint, not a wishlist.
4. Conduct a structured **PRD peer review** that surfaces specific defects rather than general impressions.
5. Revise a PRD in response to peer review and produce a **review-resolution log** that shows which feedback was adopted, deferred, or rejected, with reasoning.

Daily map

DAY	TOPIC	KEY ARTIFACT PRODUCED
1	Drafting Technical PRDs	PRD §§1–3 (context, problem, goals, scope)
2	Writing granular Acceptance Criteria	8–12 testable ACs
3	Documenting Non-Functional Requirements	NFR section across 5 categories
4	Mini-capstone — PRD assembly (non-AI)	Complete PRD ready for review
5	PRD Review — morning reviews, afternoon revisions + secondary	Revised PRD + review-resolution log

The artifact is cumulative: each day's output is added to the PRD until Friday's revised version ships.

Daily cadence (applies Mon–Thu; Friday differs)

CLOCK	BLOCK	MIX
09:00 – 09:15	Opening & objectives	Instructor
09:15 – 10:30	Teaching Block 1 + Activity 1	~25 min teach / 50 min triad work
10:30 – 10:45	Break	
10:45 – 12:00	Teaching Block 2 + Activity 2	~25 min teach / 50 min triad work
12:00 – 13:00	Lunch	
13:00 – 14:15	Teaching Block 3 + Activity 3	~25 min teach / 50 min triad work
14:15 – 14:30	Break	
14:30 – 16:00	Teaching Block 4 + Activity 4 + Wrap	~25 min teach / 45 min triad / 20 min wrap

Friday cadence (PRD review day)

CLOCK	BLOCK	MIX
09:00 – 09:15	Opening; review pairings posted	Instructor
09:15 – 10:45	Primary review round 1	Two reviewing triads on each PRD
10:45 – 11:00	Break	
11:00 – 12:00	Author response + clarifying conversations	Author triads receive feedback
12:00 – 13:00	Lunch	
13:00 – 14:30	Revisions	Authors revise based on review
14:30 – 14:45	Break	
14:45 – 15:45	Secondary review	Different reviewers; read the revision
15:45 – 16:00	Wrap + sign-off	Final PRDs + resolution logs delivered

The "non-AI" rule (Mon–Thu)

This week is the discipline of **producing a defensible written artifact without a generative tool**. The reasoning:

- AI fills in vagueness with confident generic prose. Vagueness is the bug we're trying to eliminate.
- An engineer who reads an AI-drafted PRD often spots the generic prose and discounts the document.
- The TPM muscle being built is **structured thinking expressed as written words**. AI co-writing skips the thinking.

What's allowed: spell-check, dictionary, project notes, transcripts of customer interviews, your Week-1 and Week-2 artifacts. What's not allowed: generative AI for drafting, summarizing, or critiquing any part of the PRD.

The honor system holds. Friday's reviews surface generic prose quickly.

The PRD anatomy (the same template all week)

```
# PRD – <Feature name>
**Author triad:** <names> | **Date:** <date> | **Status:** Draft / In review /
Approved

## 1. Context
Why this work, why now. Customer signal. Strategy fit.

## 2. Problem
The user's job-to-be-done; the friction they hit today. Tied to your Week-1 problem
statement and Week-2 journey map.

## 3. Goals & non-goals
- Goals: what success looks like in user terms
- Non-goals: what we explicitly will not do (often more valuable than the goals)

## 4. Scope (in / out)
The boundary. What ships, what doesn't, in this iteration.

## 5. Solution sketch
Enough description that an engineer can imagine the shape; not enough that you've
designed it for them.

## 6. Acceptance Criteria
Given/When/Then form. Granular, testable, falsifiable. (Day 2)

## 7. Non-Functional Requirements
Performance, security, accessibility, observability, compliance. (Day 3)

## 8. Metrics & validation
Tied to the Tier Sheet from Week 2. How we'll know it worked in 30 days.

## 9. Risks & open questions
Named honestly. The mark of a senior author.

## 10. Dependencies
Other teams, systems, decisions, data we need.

## 11. Out-of-scope follow-ups
Things this PRD acknowledges but doesn't ship.
```

Triads

Triads from Weeks 1–2 carry through. They authored together, they ship together. Friday's review pairings are different from authoring pairings (we cross-pollinate).

Friday review rubric

Reviewers score each PRD on:

DIMENSION	WEIGHT	WHAT "EXEMPLARY" LOOKS LIKE
Problem clarity	20%	Engineer could scope without clarifying call
AC testability	25%	Each AC is implementable + falsifiable; no "system should be intuitive"
NFR completeness	20%	5 categories present; each tied to a measurable target
Strategy linkage	15%	Goals tie to Week-2 NS / KPIs / journey friction
Risk honesty	10%	Real risks named; "no risks" is a fail
Writing discipline	10%	Clear, specific, no AI-generic prose

Bridge to Week 4

Week 4 turns from "what do we want" to "what's technically possible / constrained": monolith vs microservices, security & compliance, system mapping, latency/availability/rate-limit targets. Each triad's PRD becomes the input to the **technical architecture** conversation. The NFRs written this week are the **first draft** of the architectural constraints discussed Week 4.